The Lever Generalizes — and It Brakes: A Late, Bidirectional Action-Commitment Lever Across Agent Decisions and Architectures

Toward mechanistic circuit-breakers for irreversible agent actions

Caio Vicentino
OpenInterpretability
Fortaleza, Brazil
ORCID: 0009-0003-4331-6259
`caio@openinterp.org`

June 2026

Abstract

A prior result (*The Lever Is Late*) showed that, for a long-horizon coding agent, control of the `finish` decision lives not at the mid-layer "task is done" verdict but in a late, task-matched action-commitment block ∼30 layers downstream. That left two questions open: is the late lever *specific to termination*, and can it *brake* an action, not just elicit one? We answer both on Qwen3.6-27B over 99 SWE-bench Pro trajectories, using a second, distinct decision available in the same data — commit a file edit (`str_replace_editor`) versus continue reversible exploration (`bash`) — with n=60 deterministic decision points per condition, prefill-only patching, and generation-confirmed outcomes. **(1) Generalization (elicit).** At an explore decision, injecting a task-matched *edit*-donor into the late block makes the agent emit a real `edit` call: 0.23 → 0.77 at L59 (position control 0.08, cross-task control 0.48). **(2) The brake (suppress).** At a commit decision, injecting an *explore*-donor collapses the real edit rate 0.48 → 0.02 (96% suppression) at L55, with a same-class control intact (0.55) and the opposite-direction donor boosting to 0.92. **(3) The mechanism is monotonic and bidirectional.** Exact paired McNemar on all 14 per-point conditions yields seven contrasts that survive Holm–Bonferroni (m=7, worst p=7.6e−5); the discordance structure is the headline: elicit has c=0 (the edit-donor only *turns commits on*, never silences) and the brake has b=0 (the explore-donor only *turns them off*, never lights). The late lever moves exactly in the donor's direction with ∼zero off-direction noise. **(4) Cross-architecture.** The late-commitment *geometry* and a donor-specific late *writability* replicate in a different model family (Mistral-7B-Instruct-v0.3): the tool-commit decision is flat/negative through the first half and explodes in the late block, and a task-matched edit-donor raises P(edit) where a same-norm random vector destroys it. We frame the bidirectional late lever as the mechanism for a *mechanistic circuit-breaker*: a single late-layer intervention that blocks an action at its commit point. We demonstrate it on a state-mutating but *undoable* edit; intervening on a genuinely irreversible action (e.g. `send_transaction`) is the named next step. Tool, pre-registration, per-point data, and an adversarial pre-publication evaluation are released.

1 Background and contribution

The WANDERING arc established, across five studies on Qwen3.6-27B (reasoning-tuned, 27B parameters), a detection–control asymmetry for long-horizon agent failure: internal states that *predict* a behavior routinely fail to *control* it. The arc's first positive, *The Lever Is Late*, located control of the `finish` decision in a late action-commitment block (L51–L63), ∼30 layers downstream of the mid-layer verdict, and only when steered with a *task-matched* donor. We released `decision-locator`, a model-agnostic tool implementing three primitives — LOCATE (logit-lens per layer), SWEEP_PATCH (donor injection per layer), and STEER_GENERATE (prefill-only patch of the decision position, then free decoding). This sits in the activation-steering / representation-engineering line [6, 7], applied to a long-horizon agent's *action channel* across turns rather than a single-turn attribute.

This paper pre-registers and answers two questions that the single `finish` result could not (`PREREG_commitment_lever.md`, `PREREG_cross_model.md`). First, is the late lever a property of the *action-commitment channel* or an idiosyncrasy of termination? Second, can the lever *suppress* an action at its commit point — the mechanism a safety *circuit-breaker* would need? Our contributions:

1. The late lever **generalizes** from `finish` to a second committal action (commit an edit), by the same locate–patch–steer method, generation-confirmed (§3).
2. The same lever **brakes**: a late explore-donor patch suppresses a real commit by 96% (§4).
3. The effect is **monotonic and bidirectional** — elicit c=0, brake b=0 — with all seven pre-registered contrasts surviving Holm–Bonferroni on exact paired McNemar (§5).
4. The commitment **geometry and donor-specific writability replicate cross-family** on Mistral-7B (§6).
5. We frame this as a **mechanistic circuit-breaker** and state honestly what is shown (a semi-irreversible proxy) and what is not (a genuinely irreversible action) (§8).

2 Setup

Data and decision points. We reuse the public bundle of 99 SWE-bench Pro [8] trajectories from Qwen3.6-27B (success 40, locked 39, wandering 20). The agent's action space is three tools: `bash` (3250 calls; reversible exploration), `str_replace_editor` (1101 calls; commits a file edit; semi-irreversible), and `finish` (40). A *commit decision point* is any turn poised between committing an edit and continuing exploration; we reconstruct each faithfully at the tool-choice position, deterministically (sorted traces, ≤ 3/trajectory), giving n=60 edit-decision and n=60 bash-decision points. **Patching.** All interventions use the `decision-locator` prefill-only protocol: the late-position residual is replaced once at the decision token, then the model decodes greedily; patching every step degenerates into repetition and is not used. **Fidelity gate.** Before any causal number, P(edit) at reconstructed edit points (0.474) exceeds that at bash points (0.299), confirming the states reproduce the real commit propensity (decision turns 5.2 vs 4.5 — close, weakening a positional confound; the position-matched control below addresses it directly).

3 The late lever generalizes: eliciting a second action

At a `bash` (explore) decision point we inject a task-matched *edit*-donor late-block residual and ask whether the agent emits a real `str_replace_editor` call (Table 1). The baseline emission rate is 0.23; the task-matched edit-donor raises it to 0.77 at L59. The two controls isolate the effect: a *position* control (an explore-donor at the same points) stays at 0.08 — below baseline, ruling out "any late patch elicits an edit" — and a *cross-task* edit-donor reaches 0.48, well above baseline but below the task-matched donor. As in *The Lever Is Late*, task-matching adds a real increment; unlike `finish`, the `edit` decision also carries a substantial *generic* component (cross-task $\gg$ baseline), so task-specificity is *graded per action*, not absolute.

condition (at bash decision points, n=60)	emission rate	exact McNemar vs. baseline
no patch (baseline)	0.23	—
task-matched edit-donor @L59	**0.77**	4.7e−10 (b=32, c=0)
task-matched edit-donor @L55	0.62	2.4e−7 (b=23, c=0)
position control (explore-donor) @L59	0.08	— (vs. donor: 9.1e−13)
cross-task edit-donor @L59	0.48	— (vs. donor: 7.6e−5)

Table 1: Elicit. A task-matched late edit-donor makes a stuck-in-exploration agent emit a real edit call.

4 The same lever brakes: suppressing a commit

At an `edit` (commit) decision point we inject an *explore*-donor and measure whether the real edit emission is suppressed (Table 2). The baseline edit rate is 0.48; the explore-donor at L55 collapses it to 0.02 — a 96% suppression. The controls are clean: a same-class (edit-donor) patch at L55 does *not* suppress (0.55), and the opposite intervention — a task-matched edit-donor at L59 — pushes the rate *up* to 0.92. Thus L55 is the brake point and L59 the power point (0.03 ↔ 0.92; Figure 1). A generation-confirmed, single late-layer patch blocks a real commit.

condition (at edit decision points, n=60; baseline edit rate 0.48)	edit rate	exact McNemar
explore-donor (brake) @L55	**0.02**	7.5e−9 (b=0, c=28)
explore-donor (brake) @L59	0.03	1.5e−8 (b=0, c=27)
same-class control (edit-donor) @L55	0.55	—
opposite (edit-donor) @L59	0.92	—

Table 2: Brake. A late explore-donor patch suppresses a real commit; same-class control intact.

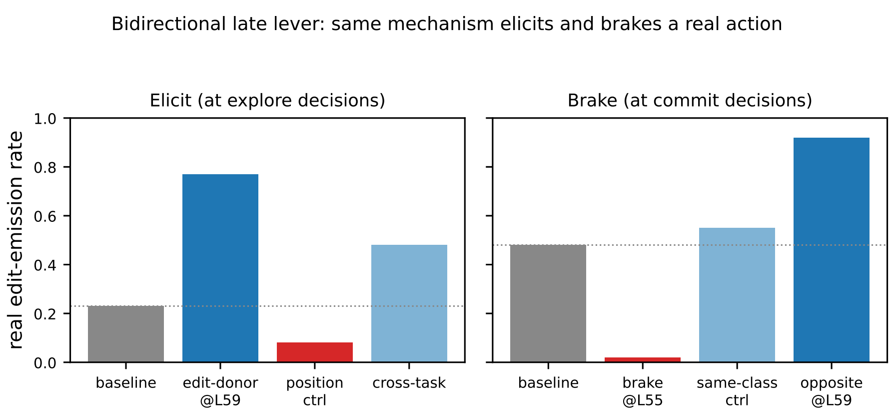

Figure 1: The same task-matched late lever elicits a real edit at an explore decision (left) and brakes a real edit at a commit decision (right), each against position / same-class and cross-task / opposite controls.

5 The mechanism is monotonic and bidirectional

We logged the per-point binary outcome for all 14 conditions and computed exact two-sided paired McNemar [10] tests. All seven pre-registered contrasts survive Holm–Bonferroni [11] (m=7; worst, task-specificity, p=7.6e−5). The discordance structure is the result: every elicit contrast has **c=0** (the edit-donor turns commits *on*, b=23–41, and *never* silences one that already fired), and every brake contrast has **b=0** (the explore-donor turns commits *off*, c=27–32, and *never* lights one). The late lever therefore moves the decision *exactly* in the donor's direction with ∼zero off-direction noise — a structural, not merely a rate-shift, claim. The one-token ΔP(edit) profile corroborates: monotonic from L23 (−0.003) to L63 (+0.685), with mid layers (L23, L31) inert and the explore-donor control going *negative* late (−0.28) — the one-token signature of the brake (Figure 2).

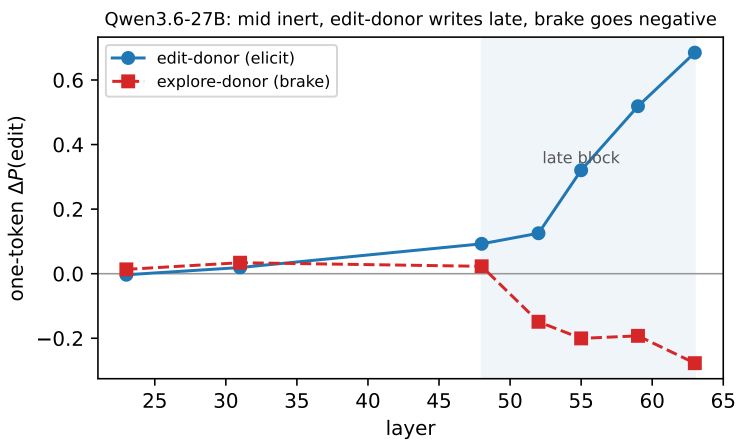

Figure 2: One-token ΔP(edit) by layer (Qwen3.6-27B). The edit-donor (elicit) is inert mid-stream and rises monotonically to L63; the explore-donor (brake) goes *negative* in the same late block. The lever is late, and bidirectional.

6 Cross-architecture replication

To test whether the late-commitment geometry is architectural rather than Qwen-specific, we ran a lite locate-plus-one-token test on `Mistral-7B-Instruct-v0.3` (a distinct model family) over 30 SWE-bench Pro tasks, with context-defined edit- vs. explore-decision points. **Geometry (robust over three runs):** the logit-lens edit gap is flat/negative through the first half (L1–L15, ∼ −2) and explodes in the late block (L19 +3.2, L23 +6.8, L27 +10.3) — the same mid-flat, late-explosion shape as Qwen. **Writability:** injecting a task-matched edit-context donor into an explore decision raises P(edit) donor-specifically (e.g. +0.15 at L27), whereas a same-norm *shuffled* vector *lowers* it (−0.61 at L27); the donor-specific margin is +0.6–0.76 in the late block. The donor is inert at L9/L11 and writes from L13 on. The qualitative pattern — a late-emerging, direction-specific, written-not-destroyed commitment representation — replicates across families (Figure 3).

Scale-matched replication (Mistral-Small-24B). To retire the "7B-only" concern, we re-ran the same lite test on `Mistral-Small-24B-Instruct-2501`, scale-comparable to Qwen3.6-27B. The result is *cleaner* than the 7B and matches Qwen more closely. *Geometry:* flat/negative through L17, exploding L19→L37 (+1.8 → +6.4). *Fidelity:* P(edit) is 0.955 at edit-context versus 0.007

at explore-context — a far sharper context separation than the 7B (0.84 vs 0.74) and comparable to Qwen. *Writability — clean mid-vs-late dissociation:* the edit-donor is *inert in the mid block* (L11/L13/L15: $-0.003/-0.002/-0.005$) and *writes almost fully late* (L29–L39: $+0.95$ to $+0.98$), donor-specific (edit-donor $+0.97$ vs. shuffled-neutral $\sim +0.5$). This is the *exact* mid-inert / late-write shape of the Qwen Tier-1 result — sharper than the 7B (which wrote from mid-depth). The late lever thus replicates across two families *and* two scales (Table 3), with the mid/late dissociation cleanest at the scale-matched 24B (Figure 4).

	Qwen3.6-27B	Mistral-7B	Mistral-24B
geometry: late-explosion	L51–63	L19–27	L19–37
fidelity P(edit) edit/explore ctx	0.47 / 0.30	0.84 / 0.74	**0.955 / 0.007**
writable window onset	late-only	mid (∼L13)	**late-only (mid inert)**
donor-specific late margin	gen-confirmed	+0.6–0.76	+0.97 vs neutral

Table 3: The late commitment lever across two architecture families and two scales. The scale-matched 24B reproduces the Qwen mid-inert / late-write dissociation most cleanly.

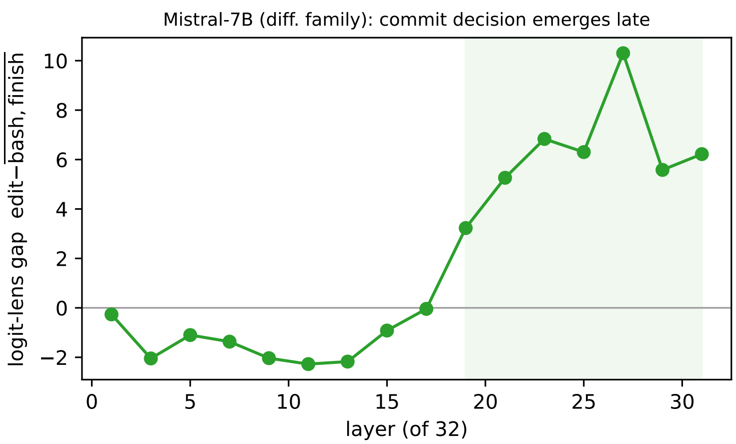

Figure 3: Cross-family LOCATE (Mistral-7B-Instruct-v0.3). The logit-lens commit-vs-explore gap is flat/negative through the first half and explodes in the late block (shaded) — the same late-emergence geometry as Qwen3.6-27B, in a different architecture.

7 Opening the lever: a sparse attention-head circuit

The lever is a single late-layer residual patch; we now open it. Because a pre-norm block writes its output as an *exact* sum, $y_L = x_L + a_L + m_L$ (input residual, attention contribution, MLP contribution), the published donor patch $\Delta_{\text{full}} = y_L^{\text{donor}} - y_L^{\text{point}} = \Delta x + \Delta a + \Delta m$ is exactly additive in the residual. We therefore re-inject each component alone with an additive hook at the layer-L output and read the one-token ΔP(edit), the same metric as the dense profile of §5. Two gates hold throughout: the additive split reconstructs the captured residual to relative error 0.0025 (L55) / 0.0024 (L59), and the `full` re-injection reproduces the published lever exactly (elicit emission $0.23 \to 0.77$@L59, brake $0.48 \to 0.03$@L55), so the decomposition measures the same intervention. All numbers are n=60/condition on Qwen3.6-27B.

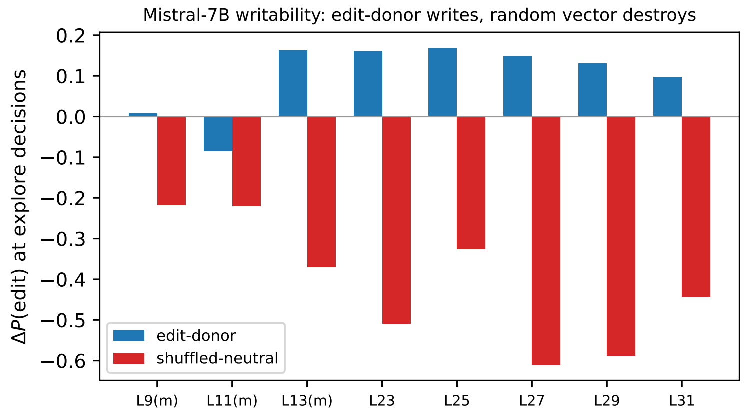

Figure 4: Cross-model writability. The task-matched edit-donor (solid) is inert mid-stream and writes the commit decision in the late block, while a same-norm shuffled-neutral vector (dashed) destroys it everywhere — the mid-inert / late-write dissociation, cleanest in the scale-matched Mistral-Small-24B.

Attention writes the elicit; the MLP is inert; the brake is distributed. Table 4 gives the recovery fraction $r = \Delta P_{\text{comp}}/\Delta P_{\text{full}}$ of each component. The two directions decompose *differently*. For the **elicit**, the L59 attention sublayer is the single largest writer (r_{attn}=0.43, 95% CI $[0.33, 0.53]$), the **MLP is null** (r_{mlp}=−0.04; paired Wilcoxon attn≫mlp p=1.8e−8), and the attention write is *direction-specific* (task-matched ΔP=+0.223 vs. random-donor +0.100, 2.2×); under generation the attention component alone roughly doubles emission (0.23 → 0.45) while the MLP component is inert (0.25). At L55 neither sublayer writes the signal — it is still *arriving* via the residual from below (r_{inp}=0.52) — so the attention write fires specifically at L59. The **brake** localizes to *neither* sublayer ($r_{\text{attn}}, r_{\text{mlp}} \approx 0$ at both layers); ∼35–40% is carried by the upstream residual and the remainder by a large negative *nonlinear interaction* (the gap $\Delta P_{\text{full}} - \sum \Delta P_{\text{comp}}$ is −0.23/−0.26). The MLP is null in all four cells. This rules out a Geva-style feed-forward "emit-action" key–value write [14]: the late commitment is an attention/residual phenomenon, and the elicit/brake asymmetry — attention-written push vs. distributed, irreducible suppression — holds at sublayer resolution (Figure 5). The strict pre-registered threshold ($r_{\text{attn}} \geq 0.5$) is not met, so we state it honestly: the elicit is attention-*dominated* but not attention-*sufficient*.

direction	layer	ΔP_{full}	r_{inp}	r_{attn}	r_{mlp}
elicit	L55	+0.319	0.52	0.05	−0.09
elicit	**L59**	+0.518	0.22	**0.43**	−0.04
brake	L55	−0.404	0.40	0.02	0.03
brake	L59	−0.382	0.35	0.00	−0.02

Table 4: Attn-vs-MLP decomposition (recovery fraction $r = \Delta P_{\text{comp}}/\Delta P_{\text{full}}$). The elicit is attention-written at L59 (MLP null, direction-specific); the brake localizes to no sublayer.

The elicit is a sparse three-head push circuit. L59 is a full-attention layer in Qwen3.6's hybrid linear/full stack (every fourth layer is softmax attention), with 24 query heads. The attention output is again an exact sum over heads, $a_L = \sum_h c_h$ with $c_h = z_h W_O^{(h)}$ [15], so we re-inject each

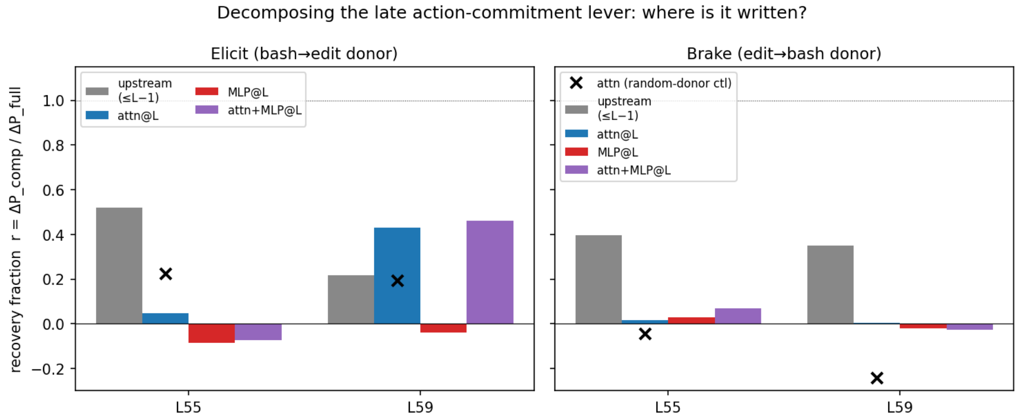

Figure 5: Component recovery of the late lever. Elicit (left) is carried at L59 by the attention sublayer (blue), MLP (red) null; brake (right) by neither sublayer. The MLP is inert in all four cells.

head's donor→point difference Δc_h alone (head-split reconstruction relative error 0.0015). The result is *sparse*: heads $8, 6, 3$ carry $+0.122, +0.119, +0.083$ while the rest are ≈ 0 or negative, and the top three together (ΔP=+0.262) *exceed* all 24 heads jointly (+0.224) because an opposing set $(16, 19, 21, \ldots,$ each -0.01 to $-0.03)$ partially cancels them — a small push–pull circuit, not a diffuse sum (Figure 6). Behaviorally the three heads alone lift emission 0.23→0.42 — 89% of the all-24-heads rate (0.47) and near the attention channel's 0.45 — and they are direction-specific (head 8 task-matched +0.121 vs. cross-repo +0.077). Geometric write-magnitude *misleads*: the largest writer along the attention direction (head 21, projection 14.0) is causally an *opponent* (−0.014) — only the patch tells the truth. The brake, re-scanned per head, has *no* head locus and is not a cancellation either (all 24 heads in $[-0.005, +0.012]$), so the elicit/brake asymmetry holds all the way down to head resolution.

head set (L59, elicit)	#heads	ΔP(edit)	edit-emit
top-1 $\{8\}$	1	+0.121	—
top-3 $\{8, 6, 3\}$	3	**+0.262**	**0.42**
all heads	24	+0.224	0.47
top head, cross-repo donor (ctl)	1	+0.077	—

Table 5: Cumulative head contribution. Three heads reproduce (and overshoot) the full attention effect; emission baseline is 0.23. Top-3 > all-24 because an opposing head set partially cancels the push.

What the heads read, and a causal source test. At the decision token the commit heads attend *globally* (low locality, ∼0.10–0.16 of mass in the last 128 keys) to the trajectory's *tool-call history* — the tool names `bash` and `str_replace_editor` and the call's JSON scaffold — not to any semantic "task done" verdict. This is the attention signature of an induction/copy mechanism [16]: the late decision promotes a tool name from the pattern of prior calls. We tested necessity by ablation [17]. The ideal head-edge knockout (recomputing $o_h = a_h V_h$) is blocked here: Qwen3.6's full attention is *output-gated* ($z_h = g_h \odot (a_h V_h)$; the ungated reconstruction fails, relerr 1.7–2.4), so we instead ablate the *source content*, mean-patching the L59 key/value at the tool-name positions.

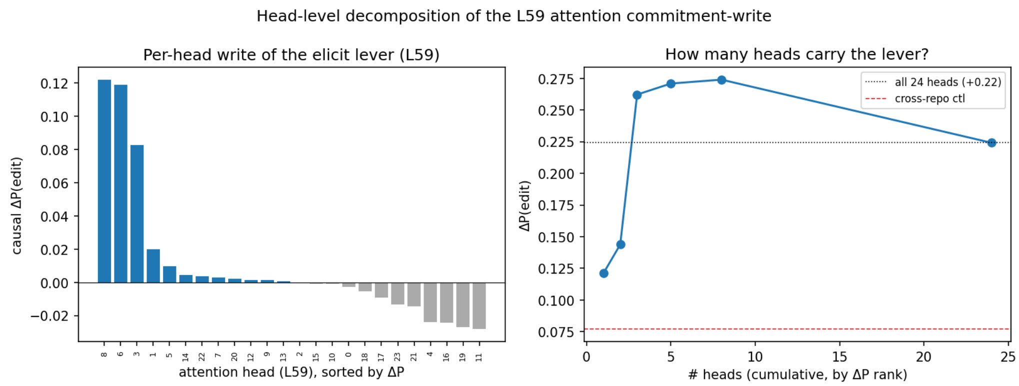

Figure 6: Head-level decomposition at L59. Left: per-head causal $\Delta P(\text{edit})$, sorted — a few heads push, a few oppose. Right: cumulative recovery; three heads reach (and exceed) the all-heads net.

The agent's tool choice is then causally specific to each tool's name tokens: ablating `bash` tokens drops $P(\texttt{bash})$ by -0.071 (-18% relative) versus -0.000 for random, count-matched positions; ablating `str_replace_editor` tokens lowers $P(\text{edit})$ and raises $P(\texttt{bash})$. Honestly, the edit side is *ceiling-confounded* — the rows that contain `str_replace_editor` tokens are already edit-saturated ($P \approx 0.9$), so its drop is small (-0.031) — and this ablation is over all heads, not head-specific. The copy reading is therefore *supported, not nailed*; a gate-aware head-edge knockout is the named next step.

Synthesis. Four levels down, the late action-commitment lever resolves to a concrete, sparse circuit: a handful of induction-style attention heads in a single late full-attention layer (L59) that read the trajectory's tool-call history and promote a tool name, partially opposed by a counter-set — not a feed-forward feature and not an attention to a semantic verdict. The brake, by contrast, has no sublayer and no head locus: it is a distributed, super-additive residual effect. The agent decides whether to commit an action largely by *copying the pattern of tools it has already used.*

8 Discussion: a mechanistic circuit-breaker, and its honest scope

The bidirectional late lever is the mechanism a safety *circuit-breaker* requires: a single intervention at an action's commit point that can block it. We emphasize what is and is not shown.

- **Generalization is an existence proof, not a universal law.** It holds for `finish` and one second action, on one model and one task family (plus a cross-family geometry/writability replication). More actions, models, and task families remain.
- **The proxy is semi-irreversible.** `str_replace_editor` mutates state but is undoable. The genuinely irreversible case (`send_transaction`, `delete`) is untested and is the named next step; the circuit-breaker is demonstrated *on a proxy.*
- **The brake redirects to exploration.** Under the brake, `bash` is the dominant action (0.60). We measured the no-brake bash baseline at edit points (0.43): the brake therefore *lifts* exploration by $+0.17$ above its already-present floor while collapsing the commit, so the suppressed decision is genuinely re-routed to reversible exploration rather than merely stalled.
- **Emission is validated by a full-call re-parse.** The elicit rate counts the tool-name prefix in a greedy continuation; to retire the onset-only concern we re-parsed the four headline

conditions for the *full* `str_replace_editor` call. The lift survives in valid calls: elicit 0.23 → 0.37 (valid-call), brake 0.40 → 0.07. Onset is thus an upper bound but the direction holds on valid calls in both directions; the brake (suppression) is the more robust to this.

These caveats were produced by an adversarial pre-publication evaluation (`EVAL_commitment_lever.md`) that recomputed every number from the released per-point data (determinism confirmed) and corrected two earlier overclaims; we release it alongside the paper.

Relation to prior work. This extends activation steering / patching [6, 7] from single-turn representations to a long-horizon agent's action channel, and operationalizes the knowledge–action gap as a *layer* gap with a *bidirectional* late lever. Our use of *circuit-breaker* is a safety-interlock metaphor and is methodologically distinct from the training-time *Representation Rerouting* of Zou et al. [12], which fine-tunes a model so harmful *content* representations become incoherent; we instead apply a single *inference-time*, late-layer patch at an agent's *action*-commitment point, with no fine-tuning. The brake direction is related to the refusal direction [13] but the steered quantity is the agent's *action commitment* (redirecting a committal action to a reversible one) rather than a generic refusal. The result is consistent with the arc's detection≠control thesis and with mechanistic-corrigibility framings of pausing before irreversible actions.

9 Conclusion

The late action-commitment lever is not specific to termination and is not one-directional: a single task-matched late patch both elicits and brakes a real agent action, monotonically and donor-specifically, with a geometry that replicates across model families. This is the mechanism for a mechanistic circuit-breaker. The next step is to demonstrate it on a genuinely irreversible action and at scale-matched second models.

Artifacts. `decision-locator` (model-agnostic), the pre-registrations, the per-point data and deterministic decision-point state, the exact-statistics script, and the adversarial evaluation are released at openinterp.org/research and on the public data bundle.

References

[1] C. Vicentino. The Lever Is Late: Causal Control of Long-Horizon Agent Termination Lives in a Task-Matched, Late Action-Commitment Block. OpenInterpretability, 2026. DOI:10.5281/zenodo.20534219.

[2] C. Vicentino. Tool-Entropy Collapse: A Cross-Architecture Signature of Agent WANDERING Failure. OpenInterpretability, 2026. DOI:10.5281/zenodo.20368600.

[3] C. Vicentino. Causal Localization of Agent WANDERING to Edge-Layer L11: The Right Locus Is Still Not a Rescue Lever. OpenInterpretability, 2026. DOI:10.5281/zenodo.20490278.

[4] C. Vicentino. Modality Matters: A Transient Behavioral Interruption Rescues Agent WANDERING Where Residual Steering Does Not. OpenInterpretability, 2026. DOI:10.5281/zenodo.20490286.

[5] C. Vicentino. The Verdict Is Not the Lever: An Interpretable Task-Completion Feature Predicts but Does Not Cause Long-Horizon Agent Termination. OpenInterpretability, 2026. DOI:10.5281/zenodo.20532769.

[6] A. M. Turner, L. Thiergart, G. Leech, D. Udell, J. J. Vazquez, U. Mini, M. MacDiarmid. Activation Addition: Steering Language Models Without Optimization. arXiv:2308.10248, 2023.

[7] A. Zou et al. Representation Engineering: A Top-Down Approach to AI Transparency. arXiv:2310.01405, 2023.

[8] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, K. Narasimhan. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? ICLR 2024. arXiv:2310.06770.

[9] nostalgebraist. Interpreting GPT: the Logit Lens. LessWrong, 2020.

[10] Q. McNemar. Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika, 12(2):153–157, 1947.

[11] S. Holm. A Simple Sequentially Rejective Multiple Test Procedure. Scandinavian Journal of Statistics, 6(2):65–70, 1979.

[12] A. Zou et al. Improving Alignment and Robustness with Circuit Breakers. NeurIPS 2024. arXiv:2406.04313.

[13] A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, N. Nanda. Refusal in Language Models Is Mediated by a Single Direction. NeurIPS 2024. arXiv:2406.11717.

[14] M. Geva, R. Schuster, J. Berant, O. Levy. Transformer Feed-Forward Layers Are Key-Value Memories. EMNLP 2021. arXiv:2012.14913.

[15] N. Elhage, N. Nanda, C. Olsson, et al. A Mathematical Framework for Transformer Circuits. Anthropic, 2021.

[16] C. Olsson, N. Elhage, N. Nanda, et al. In-context Learning and Induction Heads. Anthropic, 2022. arXiv:2209.11895.

[17] K. Wang, A. Variengien, A. Conmy, B. Shlegeris, J. Steinhardt. Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 Small. ICLR 2023. arXiv:2211.00593.